

Section - I: Multiple Choice Questions

1. What is the primary cause of underfitting in a machine learning model?
 - A) High model complexity
 - B) Insufficient training data
 - C) High bias in the model
 - D) High variance in the model

Section - II: Descriptive / Case Study-Based Questions

1. A marketing company wants to develop a machine learning model to predict the likelihood of a customer responding to a promotional email campaign based on demographic and behavioral features. They collect a dataset of 10,000 customers with 15 features, including age, location, purchase history, and email engagement metrics. After training a model on 80% of the data, they evaluate its performance on the remaining 20% and observe the following results: the model achieves an accuracy of 95% on the training data but only 60% on the test data. What might be the reason for this significant difference in performance, and how could the company modify their approach to improve the model's generalizability?